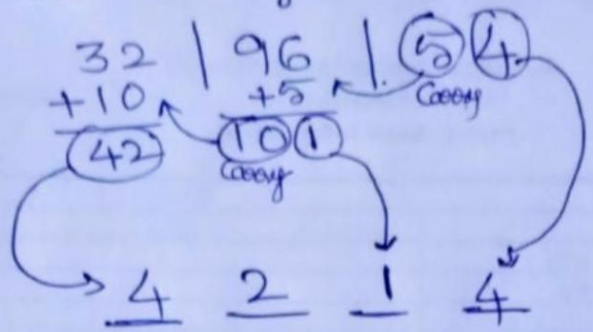
(12)

Resultant is 4 digited number. So,



one by one digit is to be placed in the vacant spaces from right to left. If any extra digits are there then carry forwarded.

(x) Product of any consecutive Numbers :-

Ex:- $32 \times 33 = 1056$

$= 32(32+1)$

$= 32^2 + 32 = 1024 + 32 = 1056$

$x \times (x+1)$
$= (x \times x) + x$
$= x^2 + x$

Product of a given number and its next number is always square of it added to it.

(xi) Finding square roots :-

↳ here we can say given number as 'x'.

↳ we have to consider as
x ends with (or) unit digit of x
x^2 ends with (or) unit digit of x^2

Note 1:- If a perfect square ends with '1' then its square root ends with '1' (or) '9'

'x' ends with	'x^2' ends
1	1
2	4
3	9
4	6
5	5
6	6
7	9
8	4
9	1
0	0

Note 2:- for '4' ℙs square root ends with '2' or '8'.

Note 3:- " '9' " " " " " '3' or '7'

Note 4:- " '6' " " " " " '4' or '6'

Note 5:- for 125 same 125.

Ex:- 1024 $\Rightarrow \sqrt{1024} = ?$

Step 1:- Given number ends with '4'. So, that ℙs square root ends with either 2/8.

Step 2:- Given number ℙs ℙn between 900 (30^2) - 1600 (40^2).

So, square root ℙs ℙn b/w 30 to 40,

means either 32/38

Step 3:- Observe 1024 ℙs near to 900 (30^2) (or) near to 1600 (40^2). ℙs near to 900 (30^2).

So, 1024 square root ℙs near to 30 & ends with 2/8 ℙs 32.

$\sqrt{1024} = 32$.

(X) Finding cube root to a perfect cube:-

x ends with	x^3 ends with
1	1
2	8
3	7
4	4
5	5
6	6
7	3
8	2
9	9
0	0